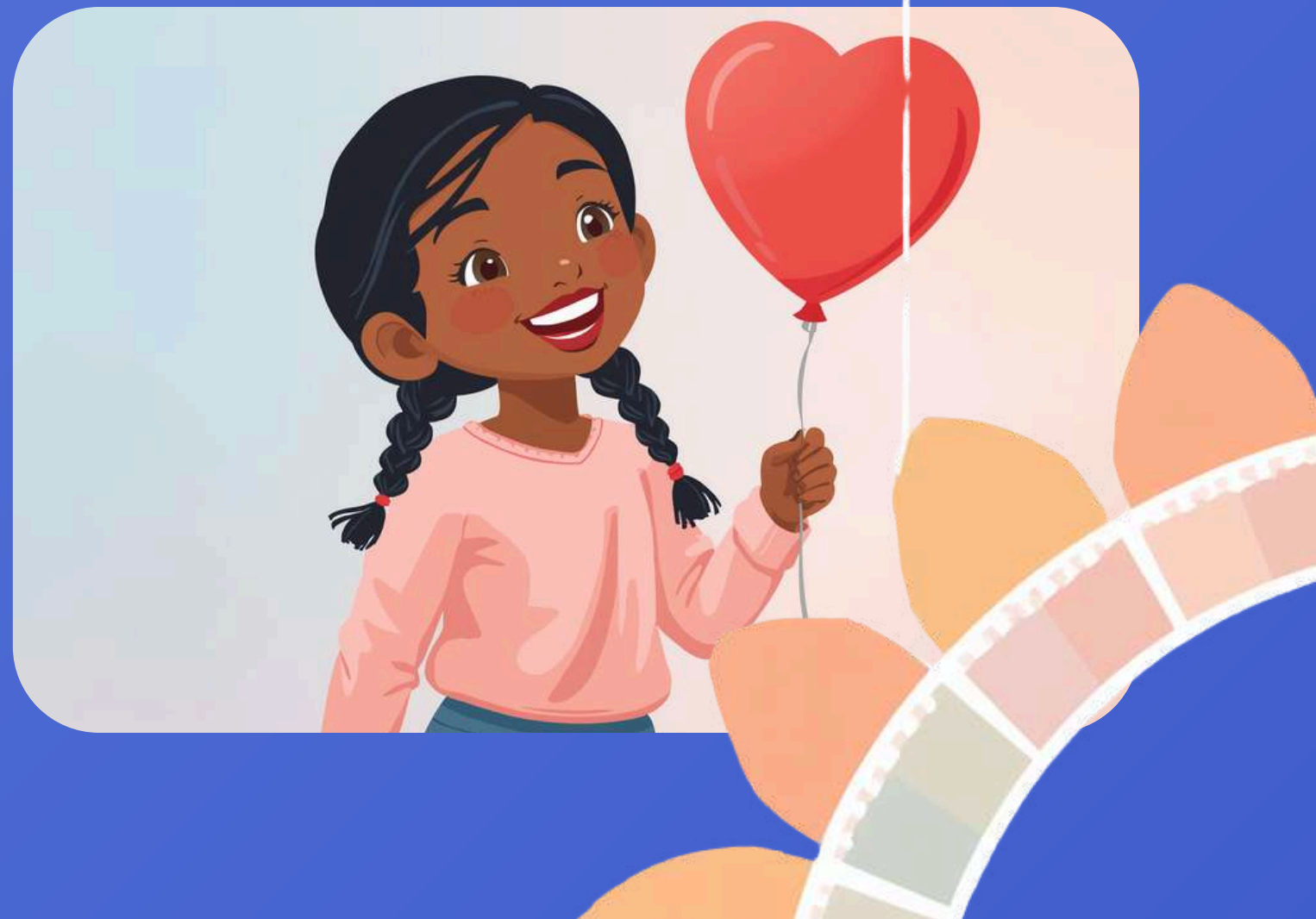


The Human Circulatory System

Discovering How Our Heart Keeps Us Alive





Why is the Circulatory System Important?

Your circulatory system is like a super delivery service that works inside your body 24/7!

Transports Oxygen: Carries fresh oxygen from your lungs to every cell in your body so you can breathe and stay energized.

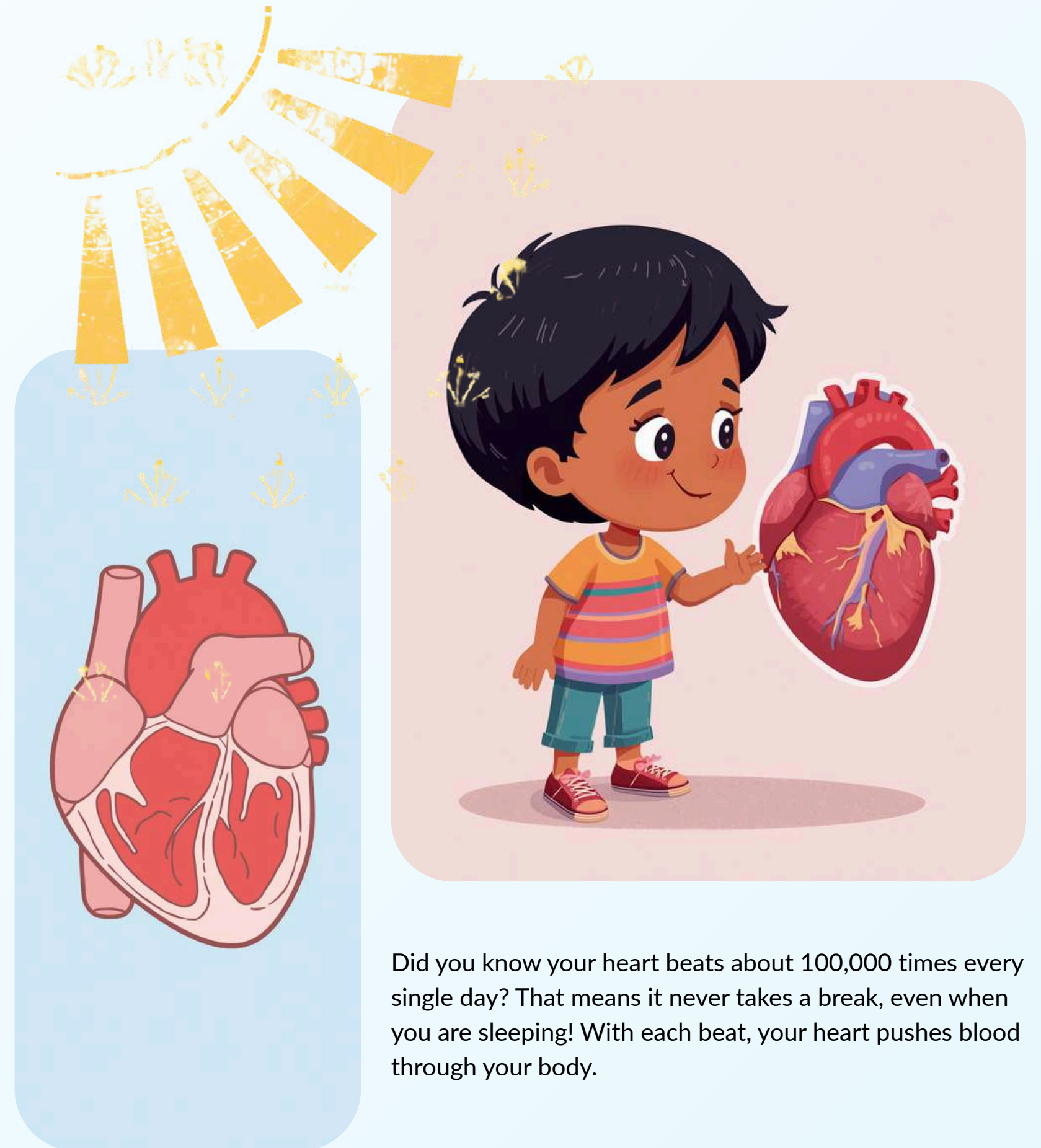
Delivers Nutrients: Brings vitamins, minerals, and energy from the food you eat to all your body parts.

Removes Waste: Picks up carbon dioxide and other waste products and takes them away to be removed.

Keeps You Alive: Without your circulatory system, your organs wouldn't get what they need to work properly!

What is the Heart?

Your heart is an amazing muscle that works non-stop to keep you alive! It pumps blood throughout your entire body, delivering oxygen and nutrients to every part of you. The heart is about the size of your fist and sits in the middle of your chest, slightly to the left.

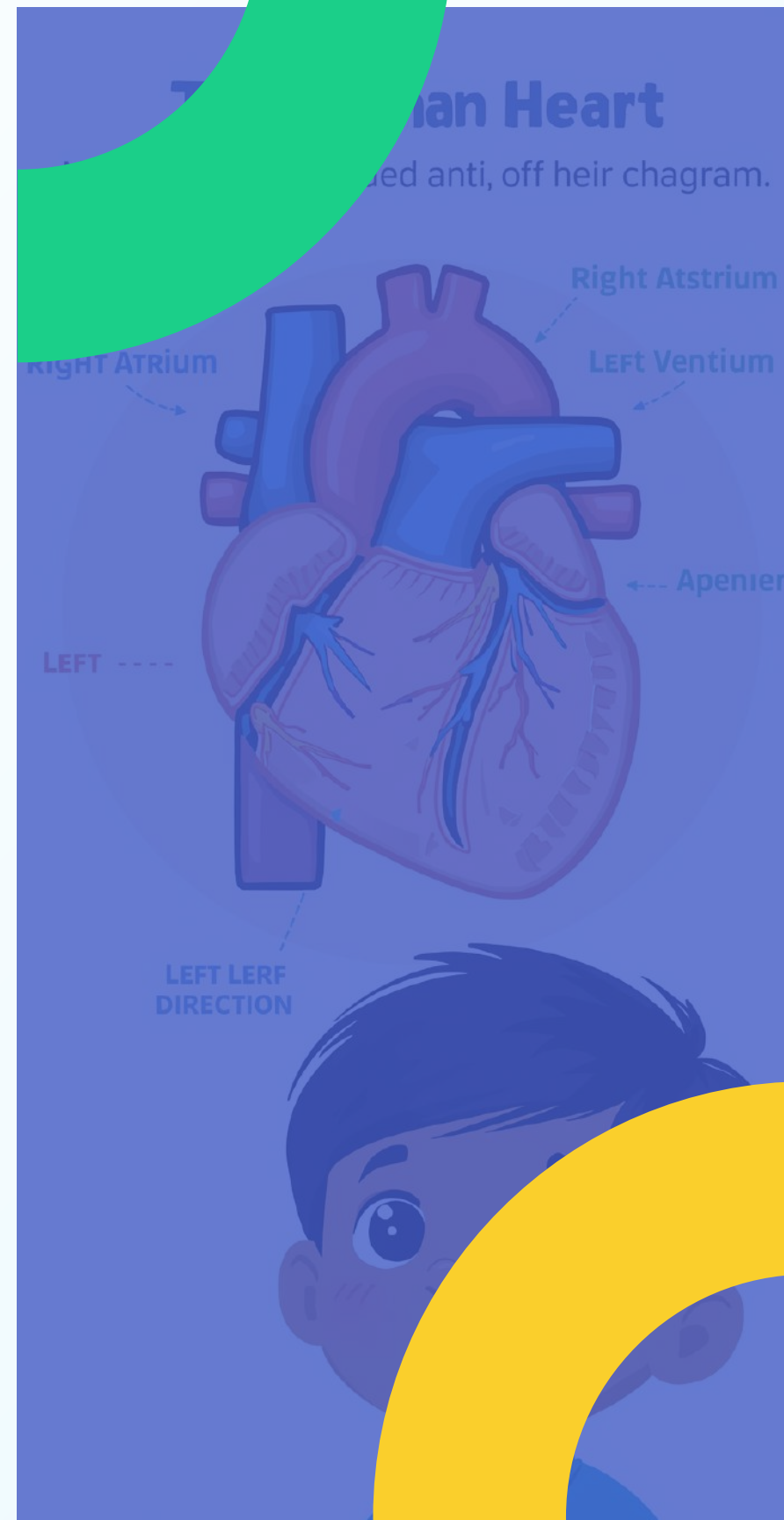


Did you know your heart beats about 100,000 times every single day? That means it never takes a break, even when you are sleeping! With each beat, your heart pushes blood through your body.



Heart Chambers and Valves

The heart has four special rooms called chambers that work together to pump blood through your body. Valves are like tiny doors that open and close to keep blood flowing in the right direction. They make the "lub-dub" sound you hear when you listen to your heartbeat!



The Four Chambers of Your Heart

Right Atrium: Receives used blood from your body that needs fresh oxygen.

Right Ventricle: Pumps blood to your lungs to pick up oxygen.

Left Atrium: Receives oxygen-rich blood returning from your lungs.

Left Ventricle: The strongest chamber! Pumps oxygen-rich blood to your whole body.

Heart Valves: Act like one-way doors to keep blood moving forward and prevent it from flowing backward.

Blood Vessels - The Highways of Blood



What Are Blood Vessels?

Blood vessels are tubes that carry blood all around your body. They connect your heart to every single part of you! Without them, oxygen and nutrients couldn't travel anywhere.

How They Work

Blood vessels are like roads inside your body! They carry blood everywhere - from your head to your toes. Arteries carry fresh blood away from your heart. Veins bring used blood back to your heart. Together, they make sure every part of you gets what it needs!



Types of Blood Vessels

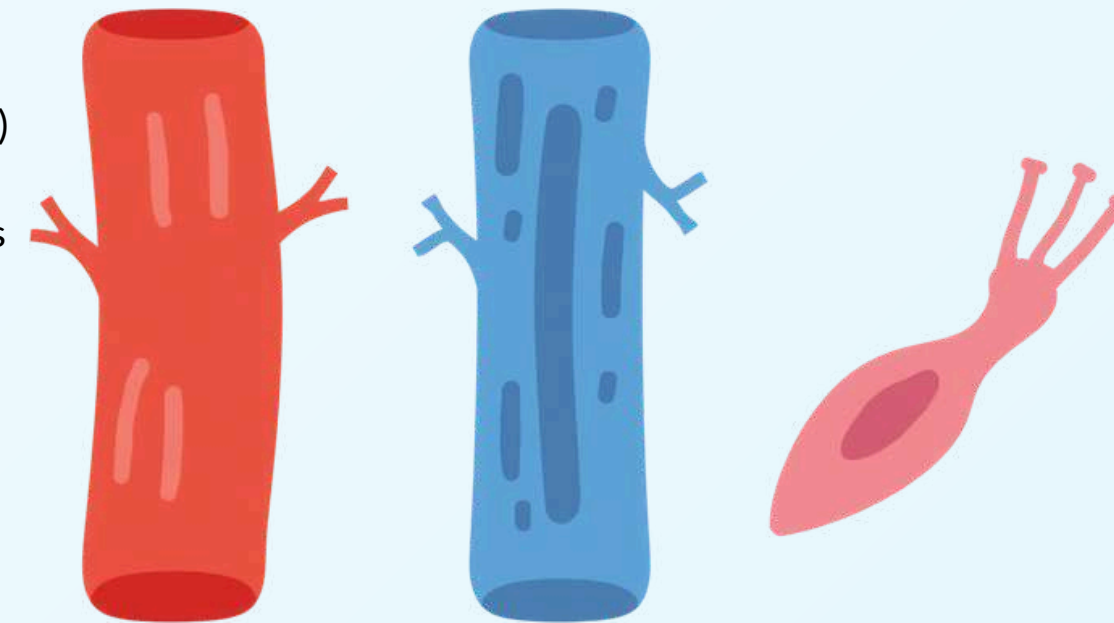
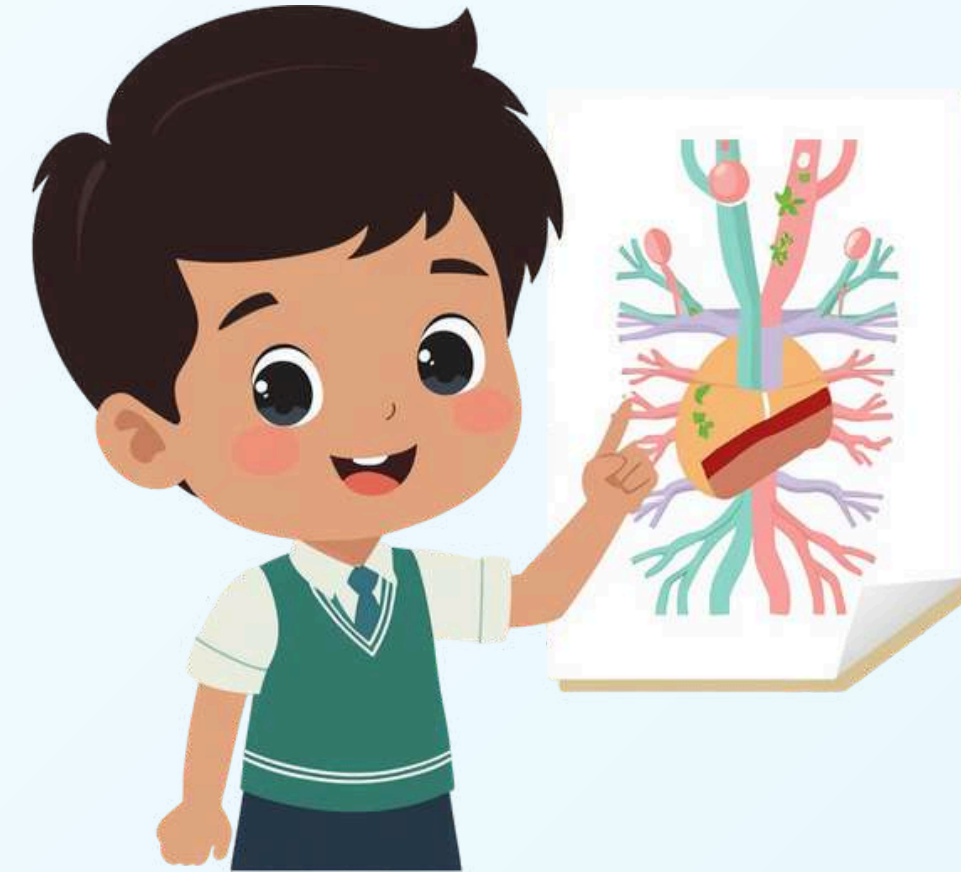
Blood vessels are like highways inside our body! They carry blood to every part of us. There are three main types of blood vessels, and each one has a special job to do. Let's learn about them!

Arteries

- Carry blood **AWAY** from heart
- Usually carry oxygen-rich blood
- Have thick, strong walls
- Pulse with each heartbeat
- Found deeper in the body
- Largest is the aorta

Veins & Capillaries

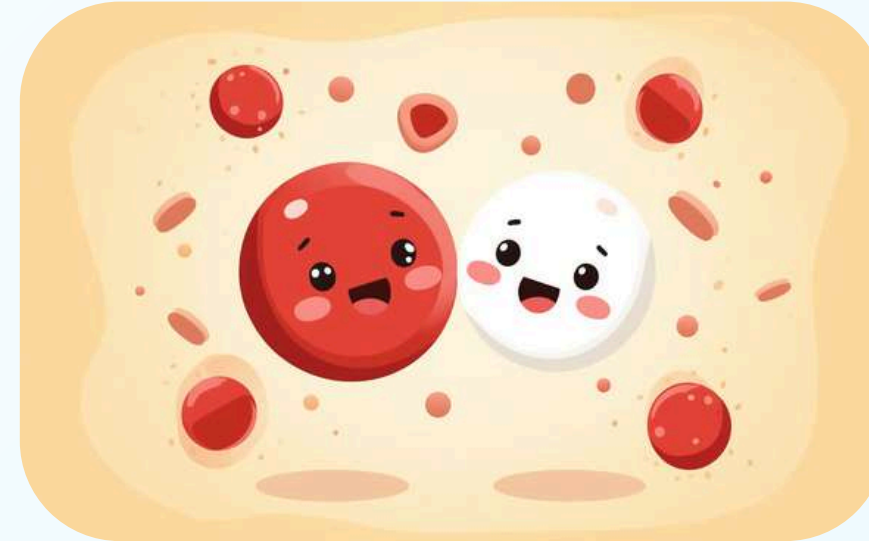
- Carry blood **BACK** to the heart
- Usually carry oxygen-poor blood
- Have thinner walls than arteries
- Contain valves to prevent backflow
- Often visible under your skin (bluish color)
- Largest are the vena cava
- Capillaries: The Tiny Connectors
- Smallest blood vessels (thinner than hair!)
- Connect arteries to veins
- Where oxygen and nutrients pass to cells
- Where waste products are picked up
- Found throughout your entire body



What Makes Up Blood?

Blood is made of four main parts that work together to keep us healthy! Red blood cells carry oxygen from our lungs to every part of our body. White blood cells are like tiny soldiers that fight germs and keep us from getting sick. Platelets help stop bleeding by clotting our blood when we get a cut.

Plasma is the watery, yellowish liquid that carries all the blood cells around. It also transports nutrients from the food we eat to all our body parts. Together, these four components make blood the amazing life-giving fluid that flows through our veins!



The Circulation Pathways

Pulmonary Circulation

Blood travels from your heart to your lungs to pick up fresh oxygen. Then it returns to the heart, ready to be pumped out to your body. This loop keeps your blood full of the oxygen you breathe in!

Systemic Circulation

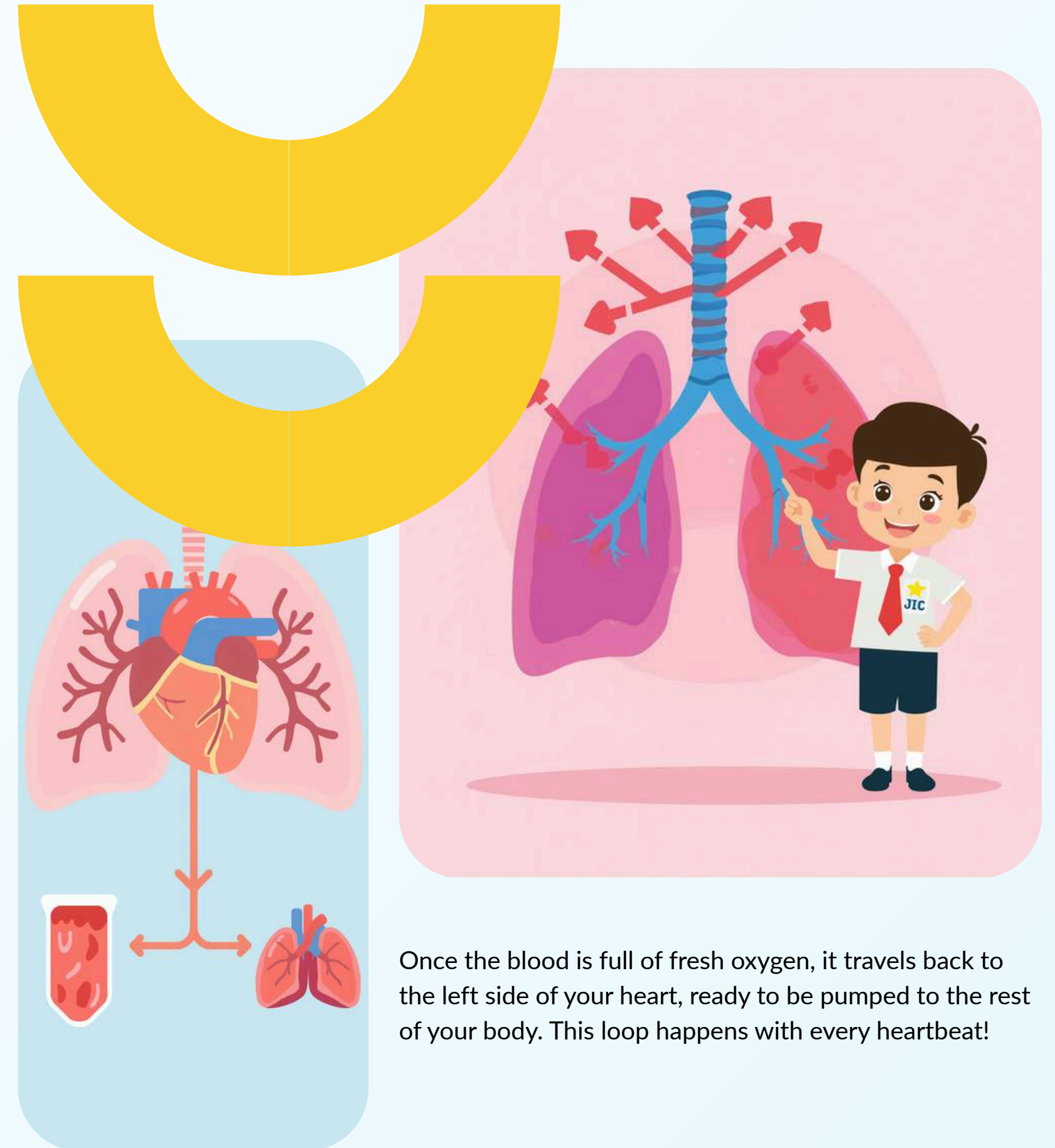
Oxygen-rich blood leaves your heart and travels through arteries to every part of your body. After delivering oxygen and nutrients, the blood returns through veins back to your heart to start again!

Lungs



Pulmonary Circulation

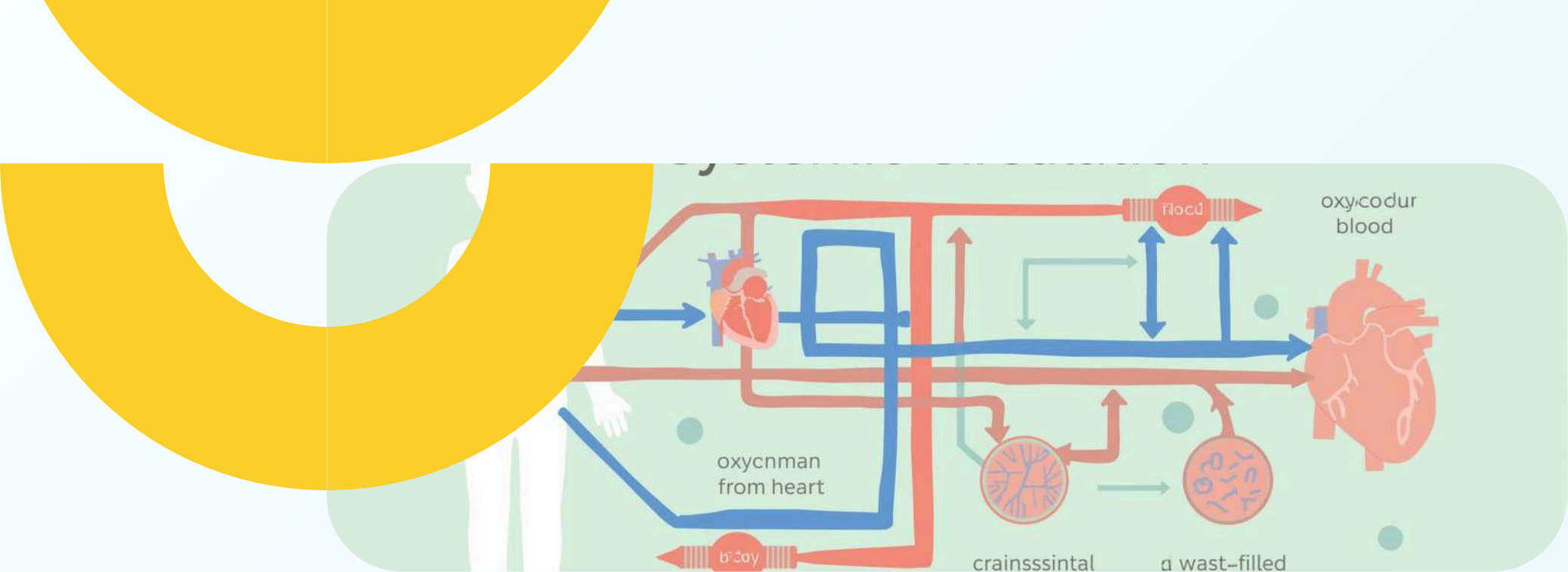
Pulmonary circulation is the special journey blood takes between your heart and lungs! When blood needs fresh oxygen, the right side of your heart pumps it to your lungs. There, the blood picks up oxygen and releases carbon dioxide (the waste gas we breathe out).



Once the blood is full of fresh oxygen, it travels back to the left side of your heart, ready to be pumped to the rest of your body. This loop happens with every heartbeat!



Systemic Circulation



How It Works:

- Heart pumps oxygen-rich blood out
- Blood travels through arteries to body parts
- Cells receive oxygen and nutrients
- Blood collects waste (carbon dioxide)
- Veins carry blood back to the heart



The Journey of Blood
Oxygen-rich blood leaves the left side of the heart through the aorta, our body's biggest artery. This blood travels to every part of your body - your brain, muscles, tummy, and even your toes! As blood delivers oxygen, it picks up carbon dioxide waste from your cells. The waste-filled blood then returns to the right side of your heart through veins.



How Oxygen and Nutrients Travel



Our blood is like a delivery truck! It picks up important things our body needs and carries them to every part of us. Let's learn how oxygen and nutrients make their amazing journey through our body!

Oxygen Transport

Oxygen enters our lungs when we breathe
Red blood cells pick up the oxygen
Oxygen attaches to hemoglobin (a special protein)
Red blood cells carry oxygen to every body part
Cells use oxygen to make energy

Nutrient Transport

We eat food that contains nutrients like vitamins, minerals, and glucose
Our digestive system breaks down food into tiny pieces
Nutrients get absorbed into the blood through intestine walls
Plasma (the watery part of blood) carries these nutrients
Nutrients travel to cells that need them for growth and repair
Extra nutrients are stored for later use by the body



Keeping Your Heart Healthy

Taking care of your heart is super important! Your heart works hard every single day to keep you healthy and strong. By making good choices about what you eat and how you move, you can help your heart stay happy for your whole life!



Tips for a Healthy Heart

Eat Healthy Foods: Fill your plate with colorful fruits, vegetables, whole grains, and proteins to give your heart the nutrients it needs.

Exercise Regularly: Run, play, dance, or swim for at least 60 minutes every day to keep your heart muscle strong and happy.

Avoid Too Much Junk Food: Limit chips, candy, and fried foods because too much salt, sugar, and fat can make your heart work harder.

Drink Plenty of Water: Water helps your blood flow smoothly and keeps your whole body, including your heart, working well.

Get Enough Sleep: Your heart rests and repairs while you sleep, so aim for 9-11 hours every night to stay healthy and energized.

Fun Facts About Your Heart and Blood



Did you know your heart and blood are amazing? Your heart pumps about 5 liters of blood every single minute! That's like filling up 5 big water bottles. And if you stretched out all your blood vessels end to end, they would reach 100,000 kilometers - that's more than twice around the Earth!



5 Liters Per Minute



100,000 km of Vessels



Beats 100,000 Times Daily





Let's Recap!

What We Learned About the Circulatory System

Your heart pumps blood through vessels, working like a delivery truck that never stops!
Blood carries oxygen and nutrients to every part of your body, keeping you energized and healthy.
Two circulation loops work together: pulmonary (to lungs) and systemic (to body).
Healthy habits like eating well, exercising, and drinking water keep your heart strong and happy!





Test Your Knowledge!

Let's see what you've learned about the circulatory system!

What does the heart do in our body?
Can you name all four chambers of the heart?
What are the two main circulation loops called?

Think about your answers and be ready to share with the class! Remember everything we learned about how blood travels through your body.





THANK YOU!

Thank you for learning about the human circulatory system with us! We hope you enjoyed discovering how your amazing heart keeps you alive and healthy. Remember, ask questions anytime - curiosity is the best way to learn!

